



Advanced Energy[®] AE SetUp for series 840, 867

**Installation and use
Operating instructions**

May 2015

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1. About these Operating Instructions

These operating instructions form part of the product.

- ⇒ Read the operating instructions carefully before installing and using the product.
- ⇒ Keep the operating instructions readily available with the device for the entire service life of the product.
- ⇒ Provide all future users of the device access to the operating instructions.
- ⇒ For more information on the device, troubleshooting, and options under:
http://www.advanced-energy.de/en/1TL_3TL_Downloads.html

1.1. SYMBOLS AND MARKUP

☑	Prerequisite
⇒	One-step instruction
1.	Multiple-step instruction
•	Bulleted list
Highlighting	Highlighting within a text
↪	Result

1.2. WARNING NOTICES

1.2.1. Layout of a Warning Notice

 WARNING TEXT	The type and source of danger are described here. ⇒ Measures for avoiding the danger are shown here.
Example	
 DANGER	Death or severe injury due to high discharge current when opening the device. ⇒ It is essential to ensure an earthing connection has been established prior to connection to the supply current circuit.

1.2.2. Categories of warning notices

There are three categories of warning notices.

 DANGER	"DANGER" designates a safety notice, the disregarding of which will lead directly to death or severe injury!
 WARNING	"WARNING" designates a safety notice, the disregarding of which can lead to death or severe injury!
 CAUTION	"CAUTION" designates a safety notice, the disregarding of which can lead to property damage or minor injury!

1.3. INFORMATION



Note:

A **notice** describes information which is important for the optimum and cost-effective operation of the equipment.

2. Safety Information

2.1. QUALIFIED PERSONNEL

The product/system related to this documentation may only be handled by **qualified personnel** for the respective scope of tasks in compliance with the documentation related to the respective scope of tasks, in particular the safety and warning notices contained in it. Qualified personnel is competent on the basis of their training and experience to detect risks and prevent potential threats while handling these products/systems.

2.2. PROPER USE OF AE-PRODUCTS

Please take note of the following:

 WARNING	Death or severe bodily injury ⇒ AE Products may only be used for the applications specified in the catalog and in the related technical documentation. If third party products and components are used, they must be recommended or approved by AE. Flawless and secure operation of the products requires proper transportation, storage, set-up, assembly, installation, start-up, operation and maintenance. The approved ambient conditions must be maintained. Notices in the related documentation must be observed.
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2.3. DISCLAIMER

We have tested that the printed text's content matches the hardware and software described. Nevertheless, deviations cannot be ruled out, and as a result, we do not assume any warranty for it matching completely. The information in this printed text will be reviewed regularly, and necessary corrections will be contained in the subsequent editions.

3. Compatibility Overview

3.1. INVERTER TYPE COMPATIBILITY

- 867 AE 3TL 8k, 10k, 13k, 17k, 20k, 23k-MV
- 840 AE 3TL 40k, 46k

4. Configuration with AE SetUp

**Note**

The AE SetUp tool is for configuring AE inverter.

**Note**

Depending on the active country code the displays can be different.

4.1. SYSTEM PREREQUISITES

The following minimum prerequisites apply for the installation and use of AE SetUp:

- Windows XP SP3 or higher
- Microsoft .NET Framework 4.0

4.2. PREPARATORY MEASURES

Double click on the file that ends with “.exe” to install AE SetUp. The installation will then be executed.

A shortcut will be saved in the program list. You can start the program via this shortcut (Start → Program → AE SetUp).

4.3. DESCRIPTION OF FUNCTIONS

You can set the country-specific parameters of the inverter with the AE SetUp tool. This covers in detail:

- Feed in conditions
- Power ramp
- Reactive power
- Actual voltage monitoring
- Actual frequency monitoring
- Phase voltage monitoring
- Average voltage monitoring
- Power reduction
- Fault ride through
- ROCOF



Note

Only the parameters enabled in the country code set for the inverter may be set. Unused parameters, e. g. additional levels of phase conductor monitoring or an undervoltage average, are saved in gray and cannot be set.

The values you should use will be preset by your distribution grid operator. If you set other values, the grid operator may take away your system's operating permit!

4.4. CONNECTION SET-UP ETHERNET AND RS485

Ethernet:

1. Set your PC's network settings to the following values: (TCP/IP)
 - IP address: **192.168.130.100**
 - Subnet mask: **255.255.255.0**
 - Standard gateway: **192.168.130.1**
2. Set up an Ethernet connection between your computer and the inverter
3. Start AE SetUp from the Windows start menu.

Here is how you access the inverter:

- via the Ethernet interface with the factory-set standard IP Address **192.168.130.30** and Port Number: **21062*** (if other values were already set, then use them)
- Select **0** in the "USS Address" protocol

RS485:

Connect the PC with the Inverter over RS485 by a converter. Then select the correct COM-port of the converter and type the USS address of the inverter. The inverter is factory-set to a baud rate 57600.

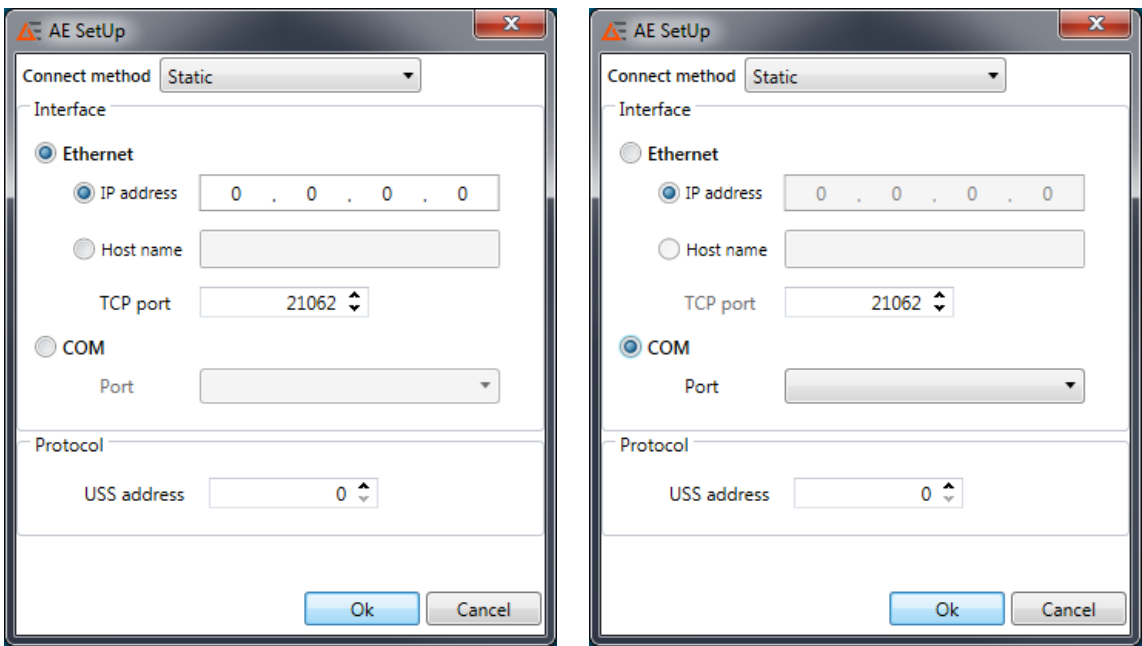


Fig. 1: Selecting the communication interface

The main window will open automatically after a connection is successfully set up and the inverter's parameter values are automatically read.

**If you have changed the communication parameters in your inverter then use them and orient toward them in regards to the TCP/IP settings.*

Scanning network areas

If you do not know the IP-address of an inverter, it is possible to scan IP ranges (e.g. If DHCP is used), subnets or entire networks for inverters.



Note

It is recommended to scan within the smallest possible IP range, otherwise you have to deal with long response times

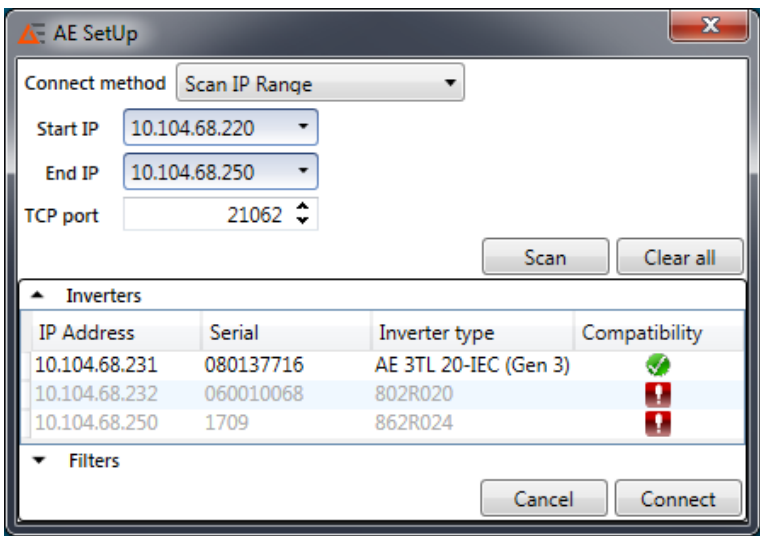


Fig. 2: Selecting the communication interface

After final scan you see in the overview list all inverters that were found in the defined area. You also see incompatible inverters; they are displayed in gray and with an icon in the compatibility column. With this version of AE SetUp-connections with incompatible inverters are not possible. For this download the corresponding AE SetUp version in the download section of the AE website.

Clicking the small triangle next to the word "filter" unfolds the filter area. Here you can filter the results for a serial number or inverter-type.

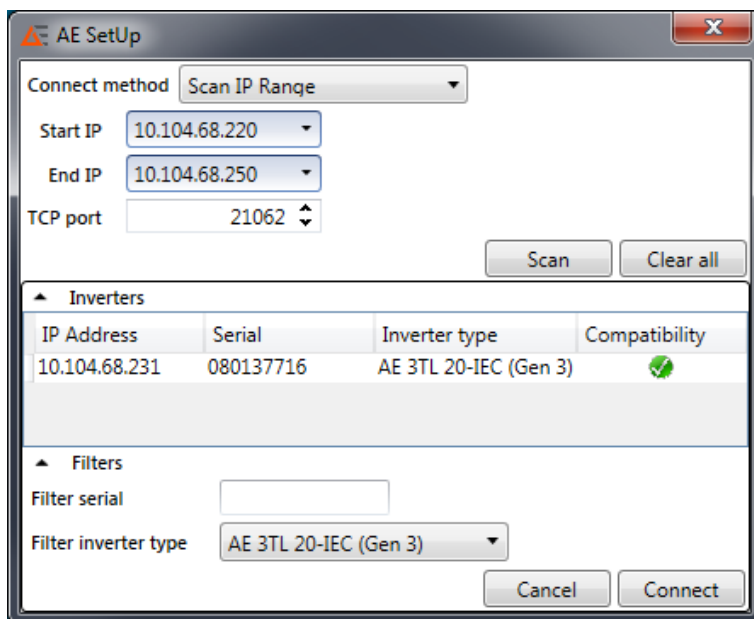


Fig. 3: Result of the scan by using the filter function

The connection to a particular inverter is carried out by double clicking on the corresponding line or clicking the button "Connect".

4.5. AE SetUp Main Window

The device information is shown in the upper part of the main window, as well as current measurement values such as line voltage and DC current in the corresponding categories: AC, DC, insulation and power. Via a pull-down menu can be selected, which current data is displayed. The choices are AC, DC, sensors are (temperature and radiation), energy and feed-in (reactive power requirements and power reduction targets).

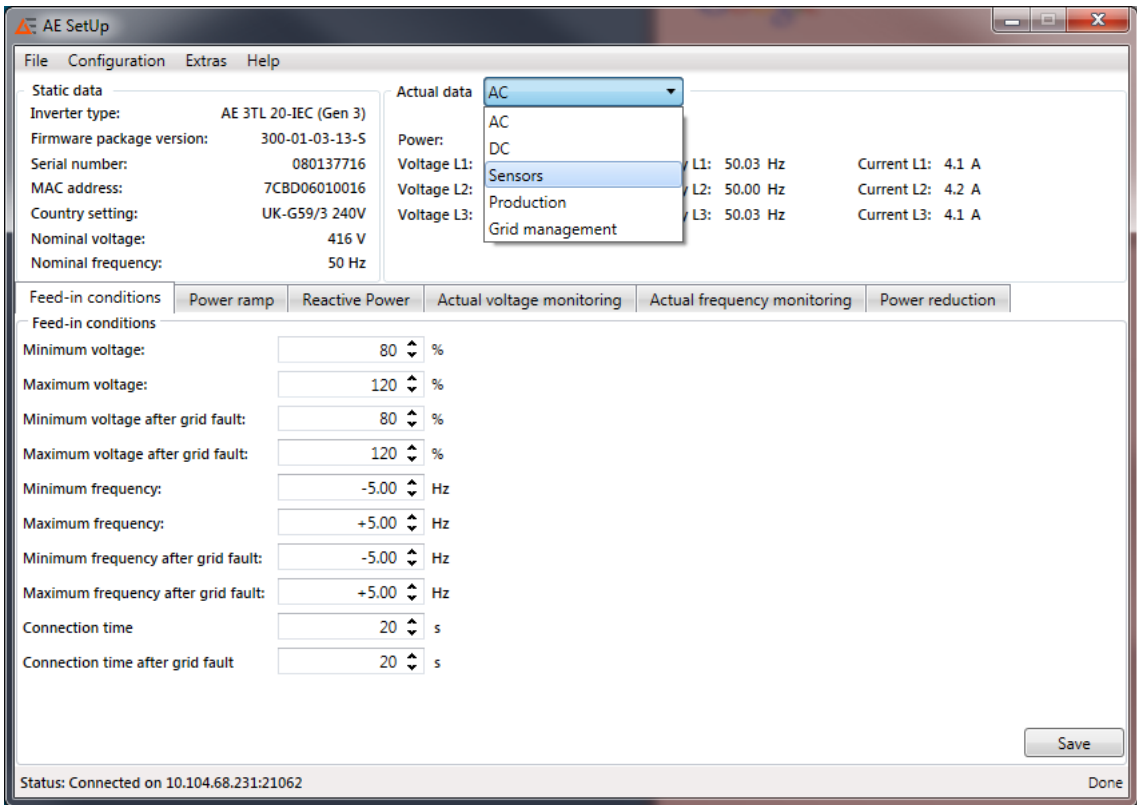


Fig. 4: Main Window AE SetUp

In the lower part, you will find options for settings sorted in tabs. Different tabs are selectable depending on the country configuration selected

To change the inverter's limit values of the country a password is needed. This is available after completing the Confidentiality Agreement (found in the downloadable zip file on the AE website) on request to the AE Services.

To set the password click on "File → Settings → Passwords".

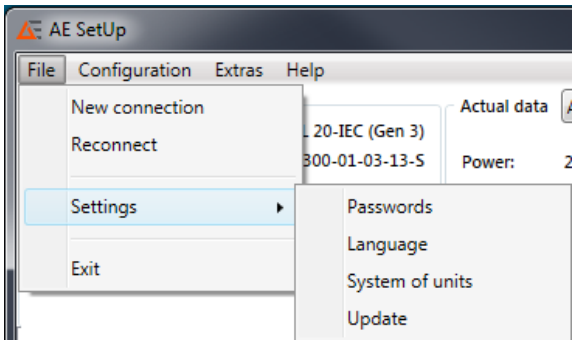


Fig. 5: Selection settings



Note

Changed values in all screens described below are only active when the Save button is pressed.

**Note**

Voltage and frequency limits are always shown relative to the nominal value. Voltage values are shown as a percentage of the nominal voltage, frequency values as the difference to the nominal value.

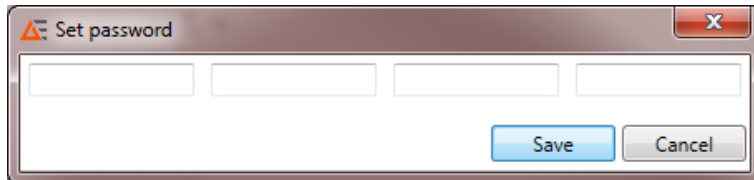


Fig. 6: Input password

4.6. Feed in conditions

You can set the inverter's feed-in conditions here, i.e. which line voltage range and line frequency range the inverter will begin to activate to supply in.

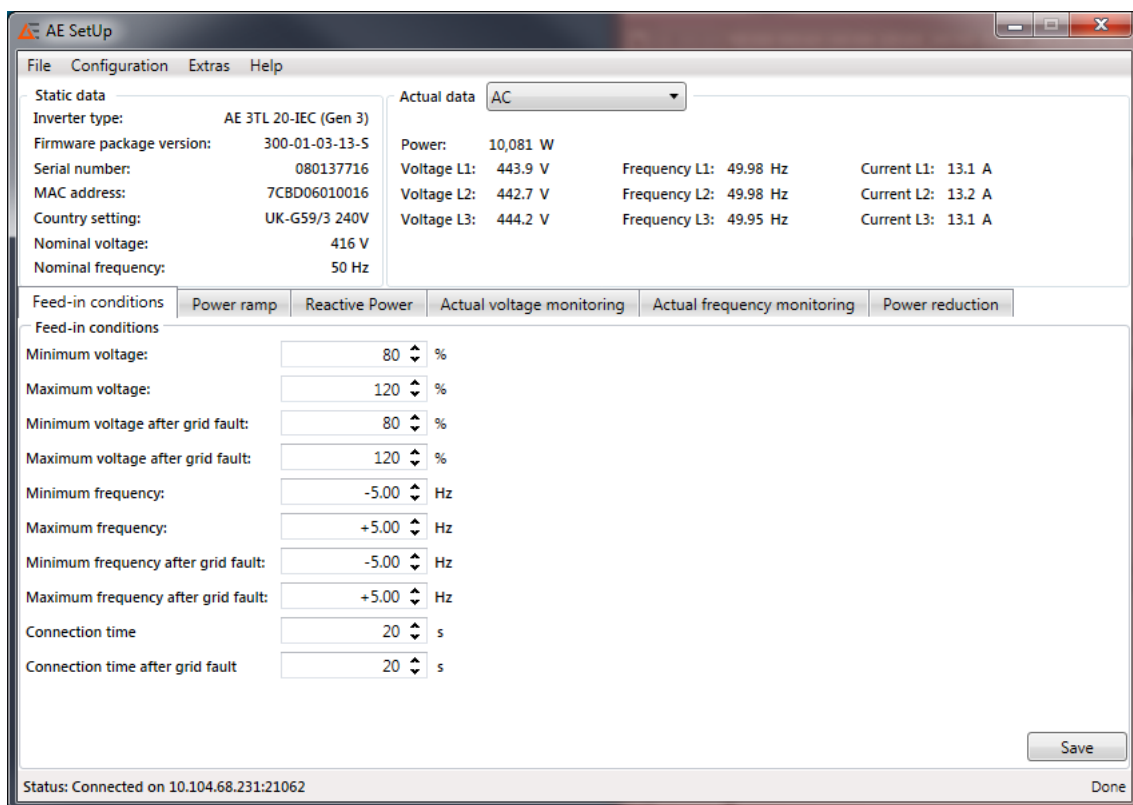


Fig. 7: Feed-in Conditions

4.7. POWER RAMP

A power ramp is required in the current standards for connecting PV systems to German electrical grids, i.e. after a grid error on the AC voltage side, the power plant is not allowed to deliver maximum real power again after getting back on the grid until after a certain period of time. The duration from power-up until reaching maximum power can be set here.

You can set a ramp time after grid connection and a ramp time for the re-up after a power failure.

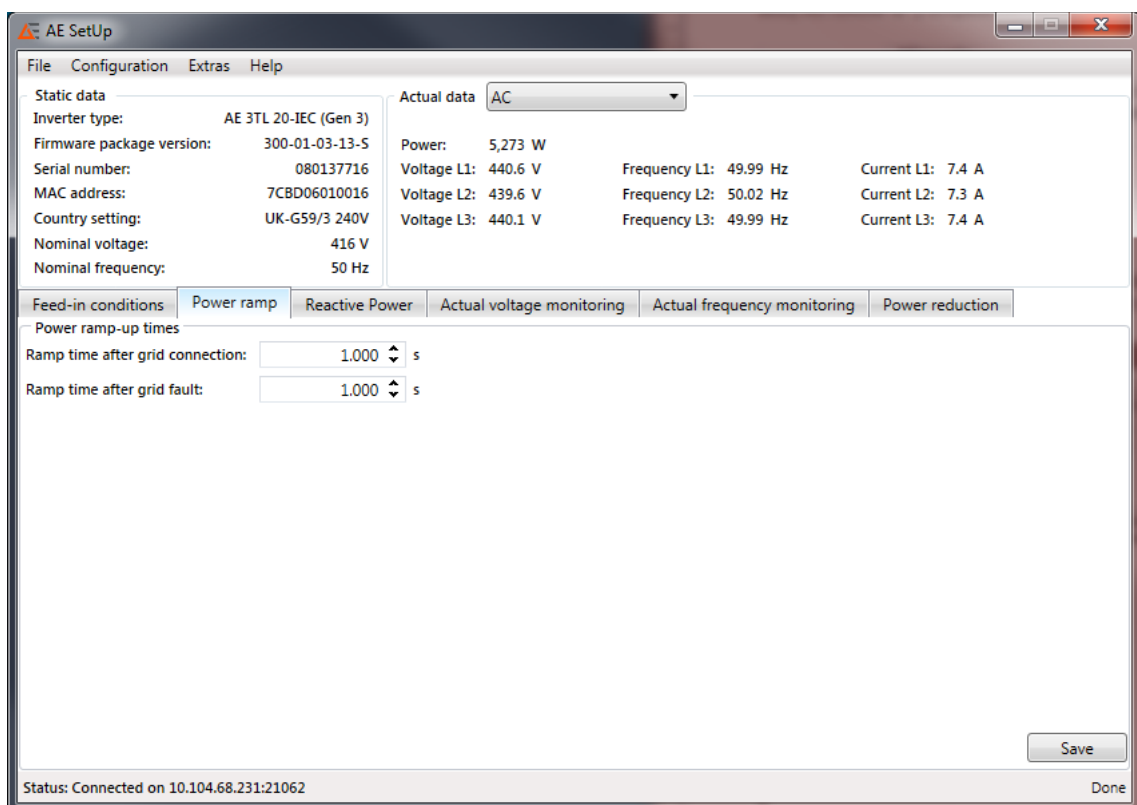


Fig. 8: Power Ramp

4.8. ACTUAL VOLTAGE MONITORING

Use the actual voltage monitoring tab to set the voltage limits and their power-down times for Y-voltage (phase - neutral).

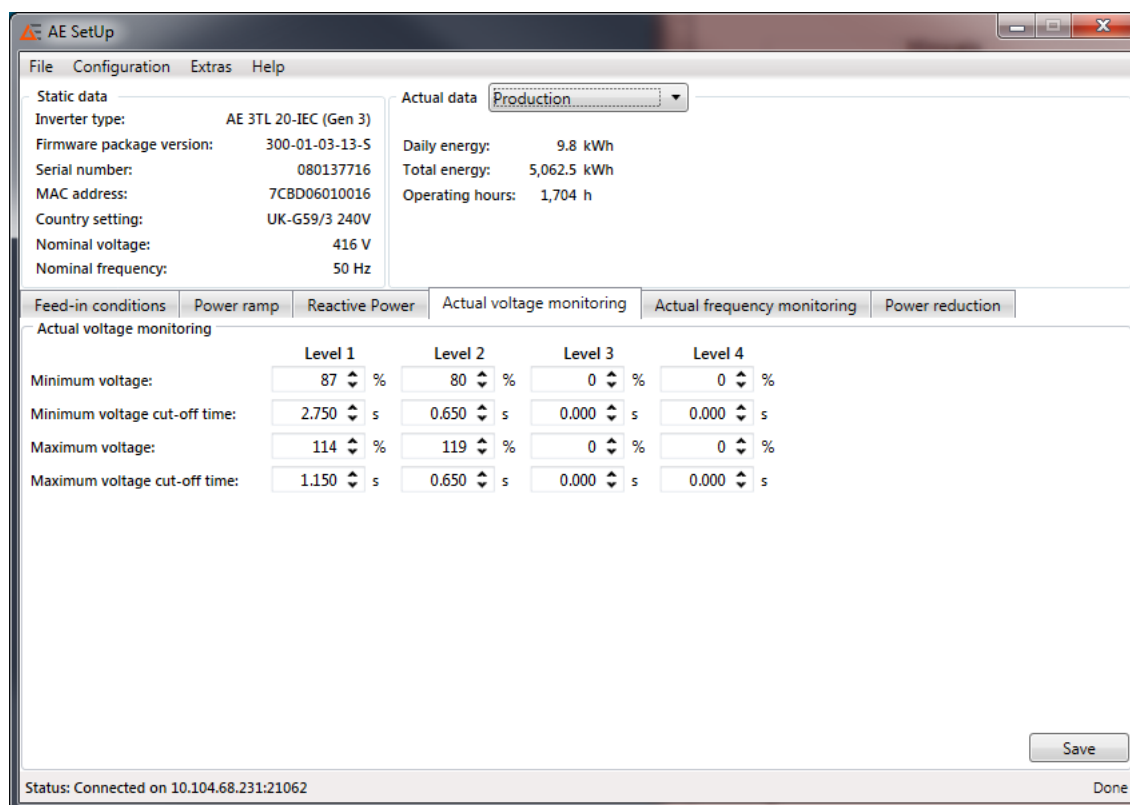


Fig. 9: Actual voltage Monitoring

4.9. ACTUAL FREQUENCY MONITORING

Use the actual voltage monitoring tab to set the voltage limits and their power-down times.

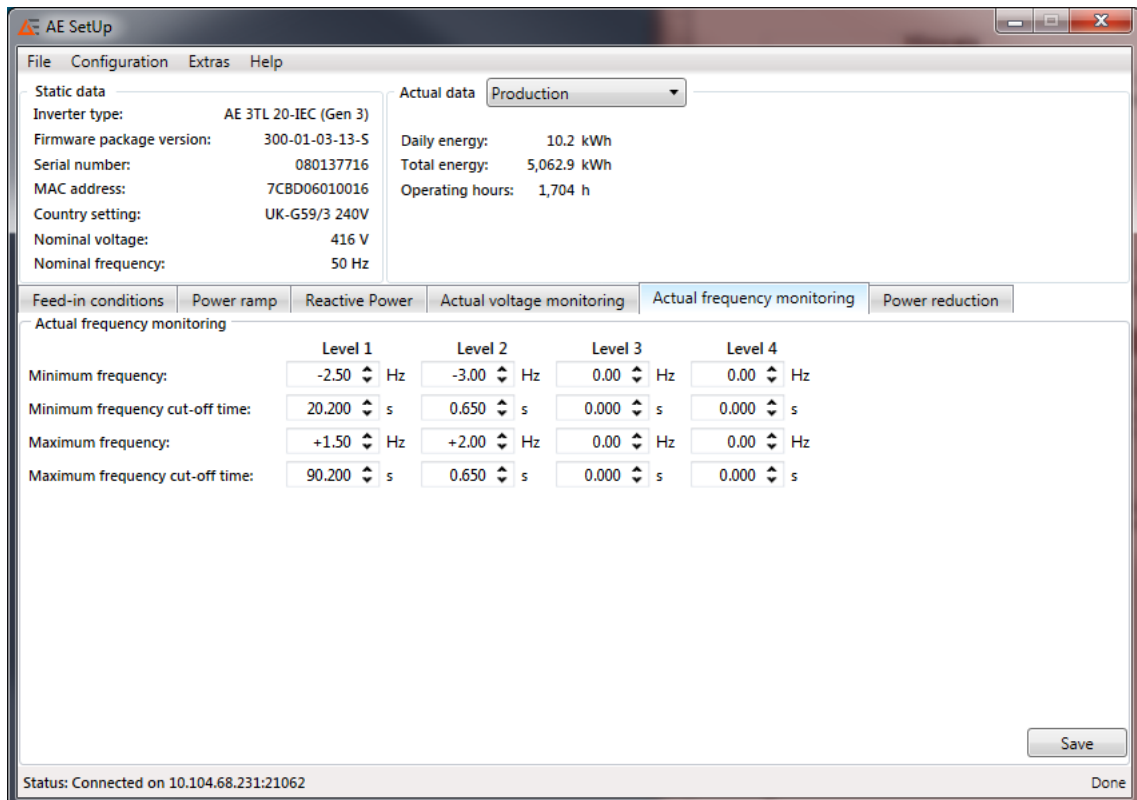


Fig. 10: Actual frequency monitoring

4.10. POWER REDUCTION

Hard power-down of PV systems in conjunction with overfrequency can put the grid's stability in jeopardy. This is why the frequency-dependent reduction was introduced. The maximum deliverable power is reduced starting with a certain frequency value depending on the frequency. You can set the start value for the reduction, the reduction factor and the return frequency in the Power reduction tab.

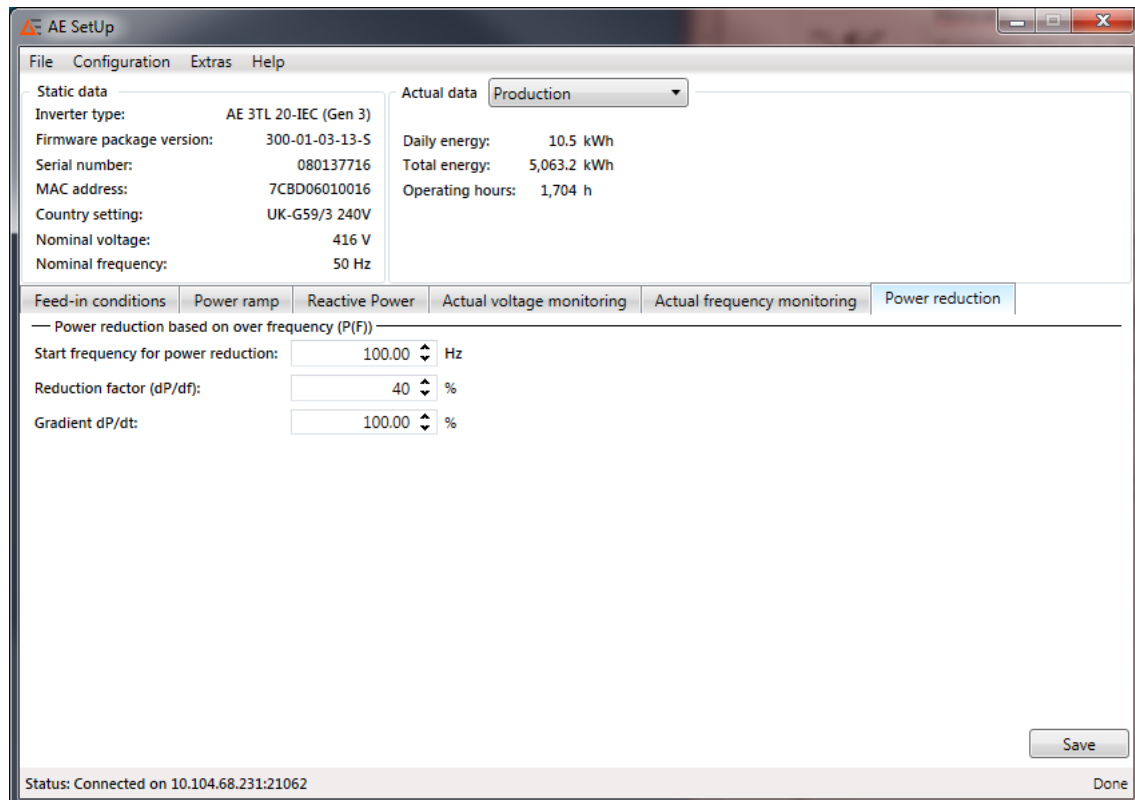


Fig. 11: Relay Configuration for Power Reduction

4.11. REACTIVE POWER

There are different modes of operation available in the Reactive power tab to enable you to play your part in regards to the voltage stability of the electrical grids as required by the electric power company.

A **Capacitive** reactive power supply is brought about by specifying a **negative angular offset (overexcited)**. The reactive power supply will accordingly be made **inductive** by setting a **positive angular offset (underexcited)**.

The phase offset of the overall system caused by the wiring and external components can be adjusted with the help of the "Device-specific offset" to enable a starting condition with $\cos \varphi = 1$ to be obtained.

This function is always active regardless of which mode of operation is selected.

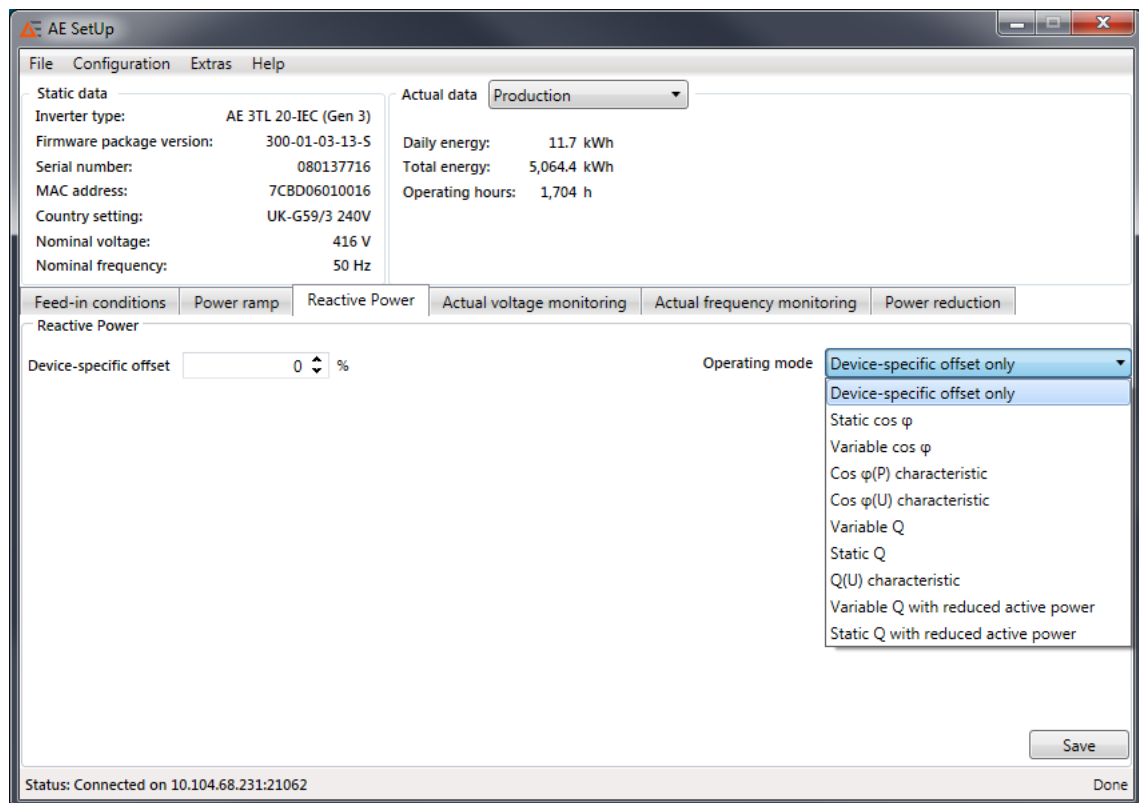


Fig. 12: Different modes of operation

4.11.1. Device-specific offset only

Only the device-specific offset can be set.

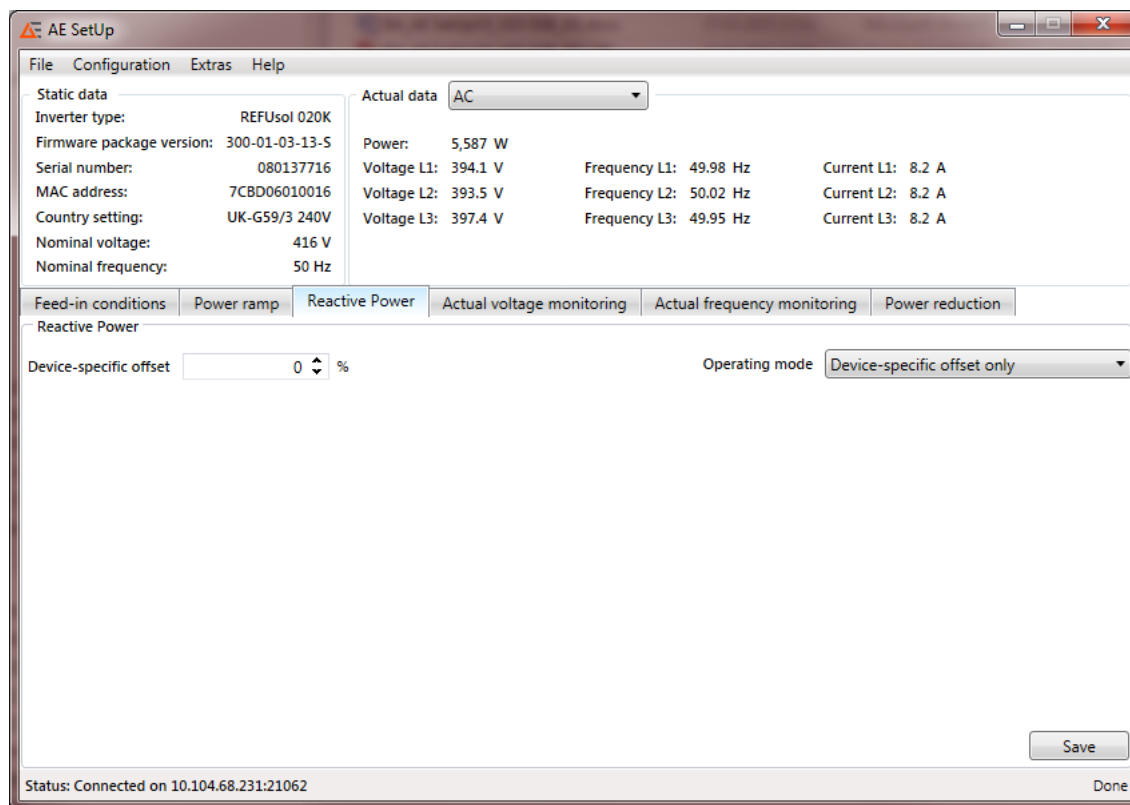


Fig. 13: Device-specific offset

4.11.2. Static $\cos \varphi$

A permanently applicable phase shift is specified via the constant, and a constant reactive power supply is thus achieved.

You can either enter the constant via a phase shift in degrees ($^{\circ}$) or via $\cos \varphi$.

The "rate of change of phase offset" indicates how fast the inverter adjusts to the new $\cos \varphi$ -fixed value.

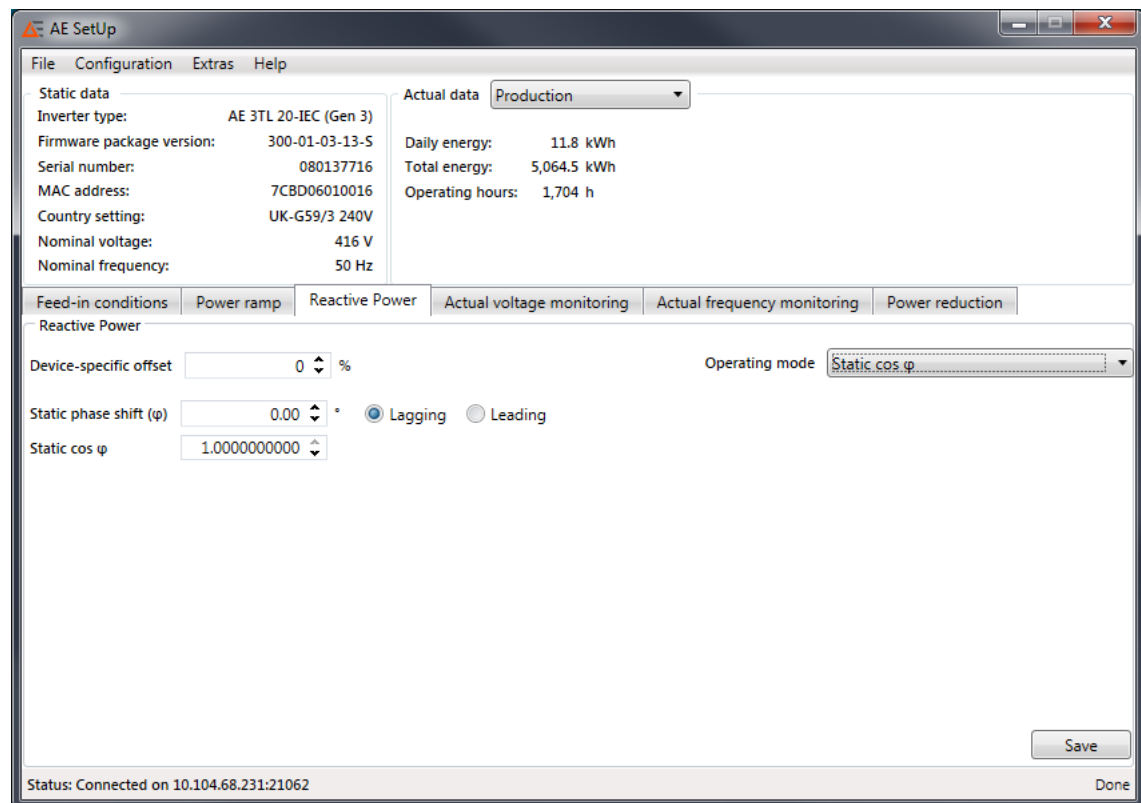


Fig. 14: Static $\cos \varphi$

4.11.3. Variable $\cos \varphi$

This mode of operation is solely functional when used with a PMU or other external compatible control units. The corresponding parameters for special adjustments of the function are stored in the PMU. In the window a device specific offset can be specified.

The value for the phase shift is shown in the window.

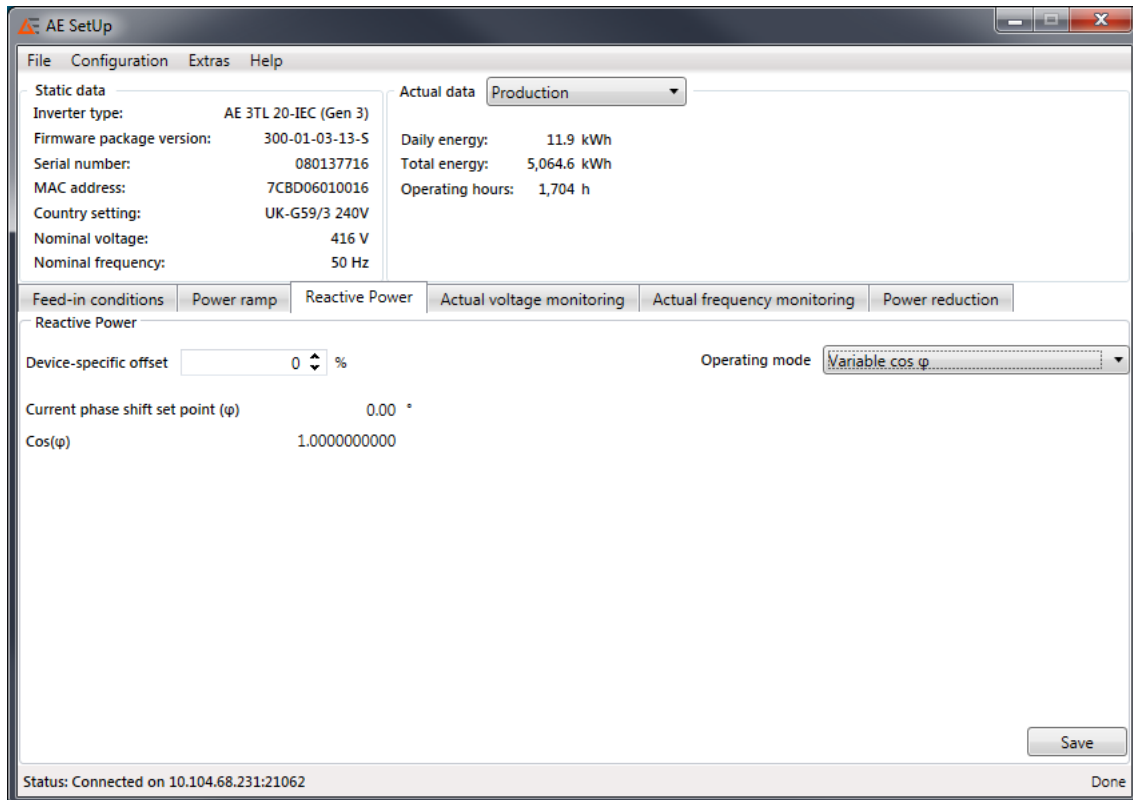


Bild 15: Variabler $\cos \varphi$

4.11.4. $\cos \varphi$ (P) - characteristic

The desired phase shift and reactive power supply can be specified in this function depending on the power supplied by the inverter.

The settings are available in the whole 0-100% power range in increments of 10%.

You can either enter the constant via a phase shift in degrees (°) or via $\cos \varphi$.

The "rate of change of phase offset" indicates how fast the inverter is adjusted to a new value.

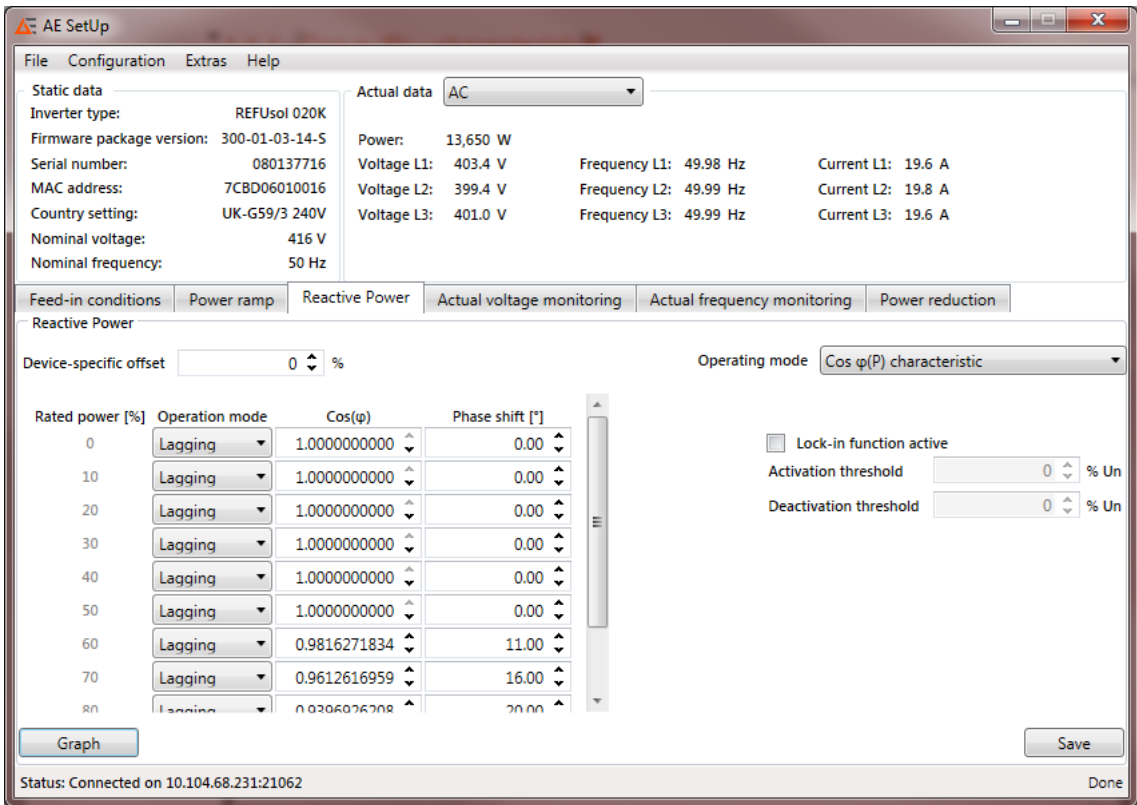


Fig. 16: Cos φ (P) characteristic

You can display the characteristic entered in a chart by pressing the "Graph" button.

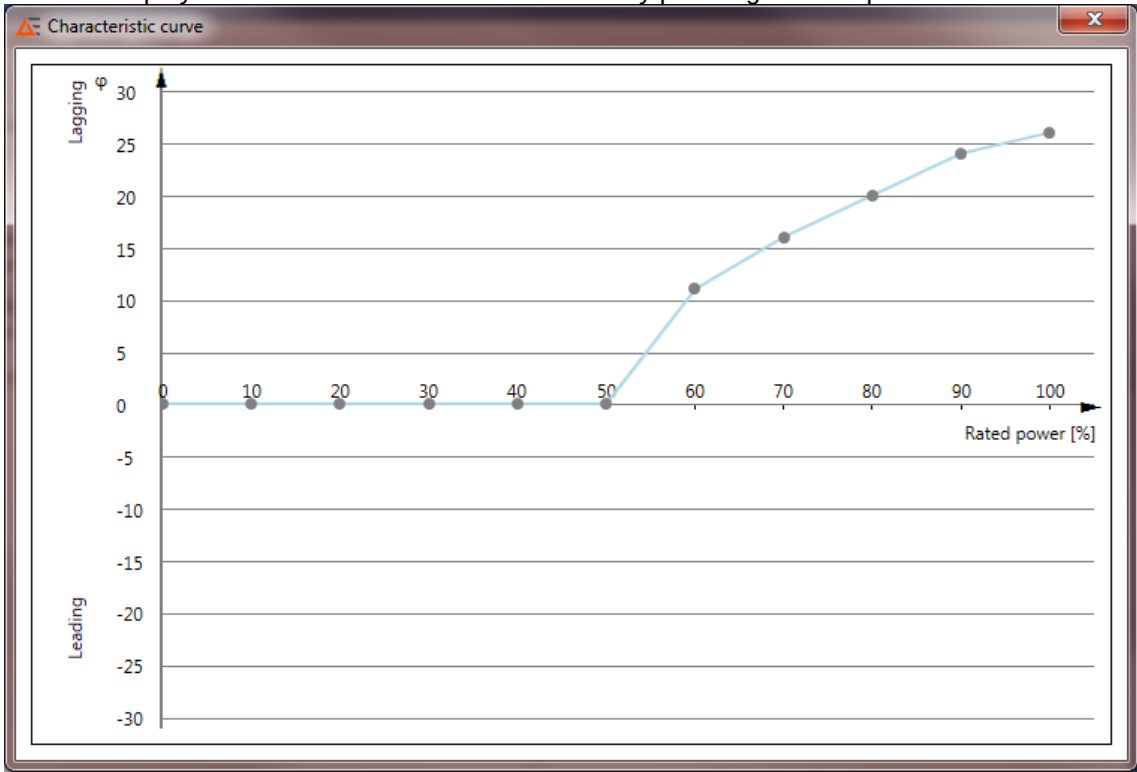


Fig. 17: Cos φ (P) – Characteristic Chart

4.11.5. $\cos \varphi$ (U) – characteristic

The desired phase shift and reactive power supply can be specified in this function depending on the current line voltage.

The settings are available in the 90-110% range in reference to the nominal voltage in increments of 2%.

You can either enter the constant via a phase shift in degrees ($^{\circ}$) or via $\cos \varphi$.

The "Time constant for phase shift" factor, in addition, indicates how fast the inverter will react to measured voltage changes with the reactive power supply.

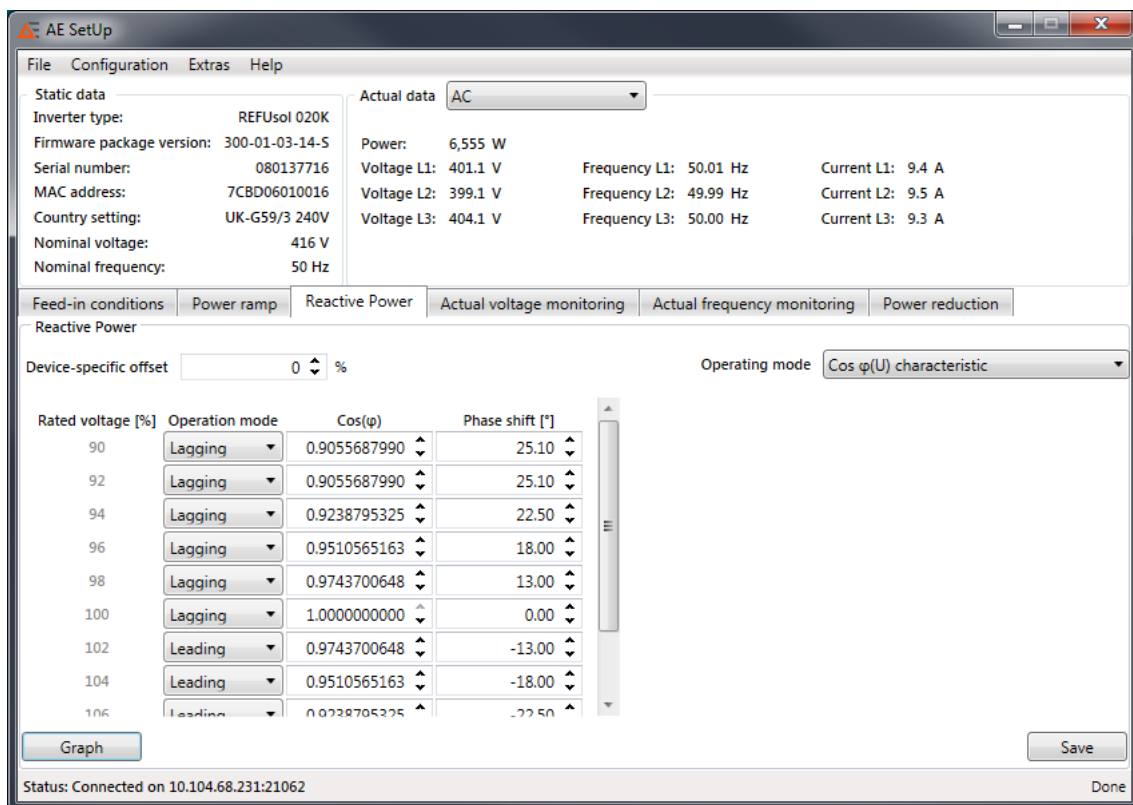


Fig. 18: $\cos \varphi$ (U) – characteristic

You can display the characteristic entered in a chart by pressing the "Graph" button.

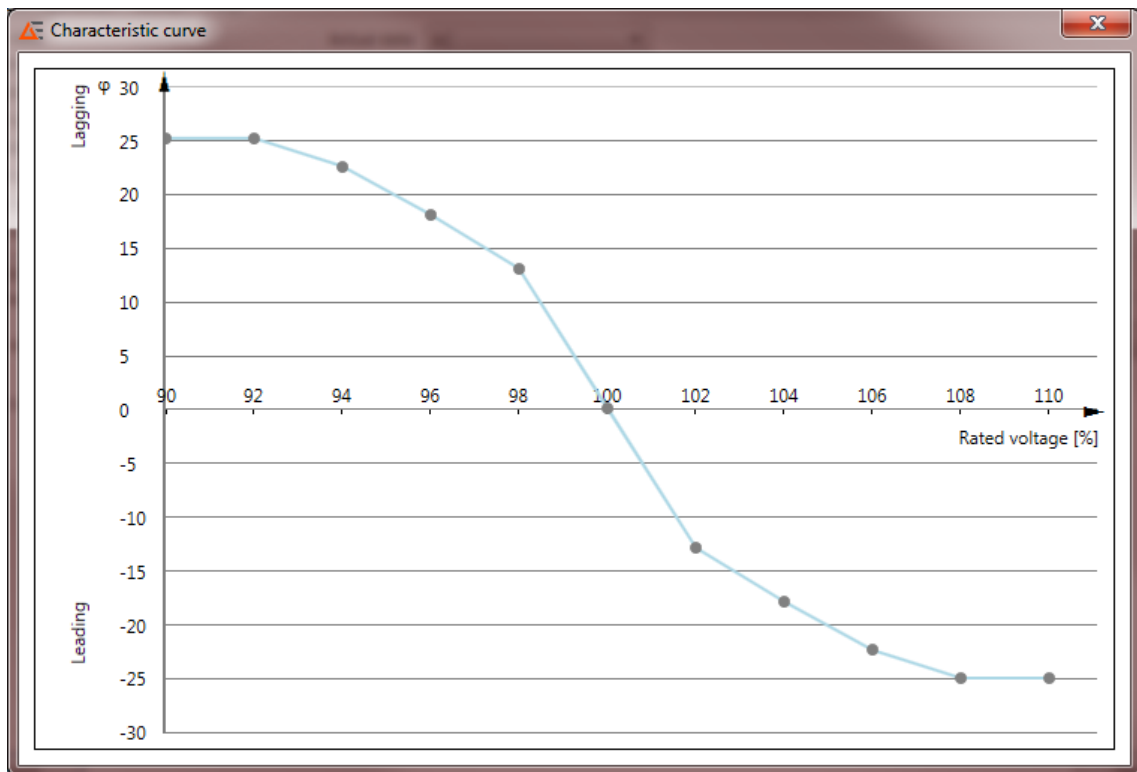


Fig. 19: Chart $\cos \varphi (U)$ – Characteristic

4.11.6. Variable Q

The Q value is determined by other external, compatible regulation and control units. In this screen only the "Device-specific offset" can be adapted.

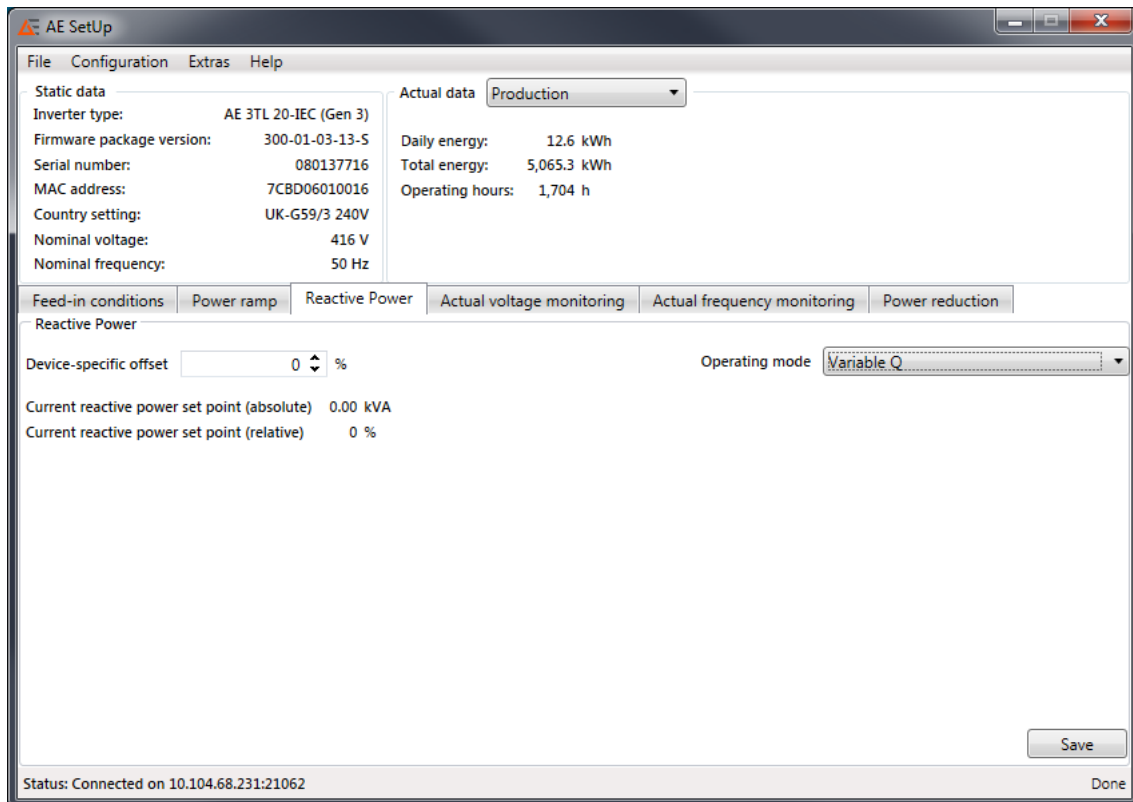


Fig. 20: Variable Q

4.11.7. Static Q

It is possible to enter a certain reactive power as fixed value. The fixed value can be entered absolutely as kVA or relative as %.

A device-specific offset can be specified.

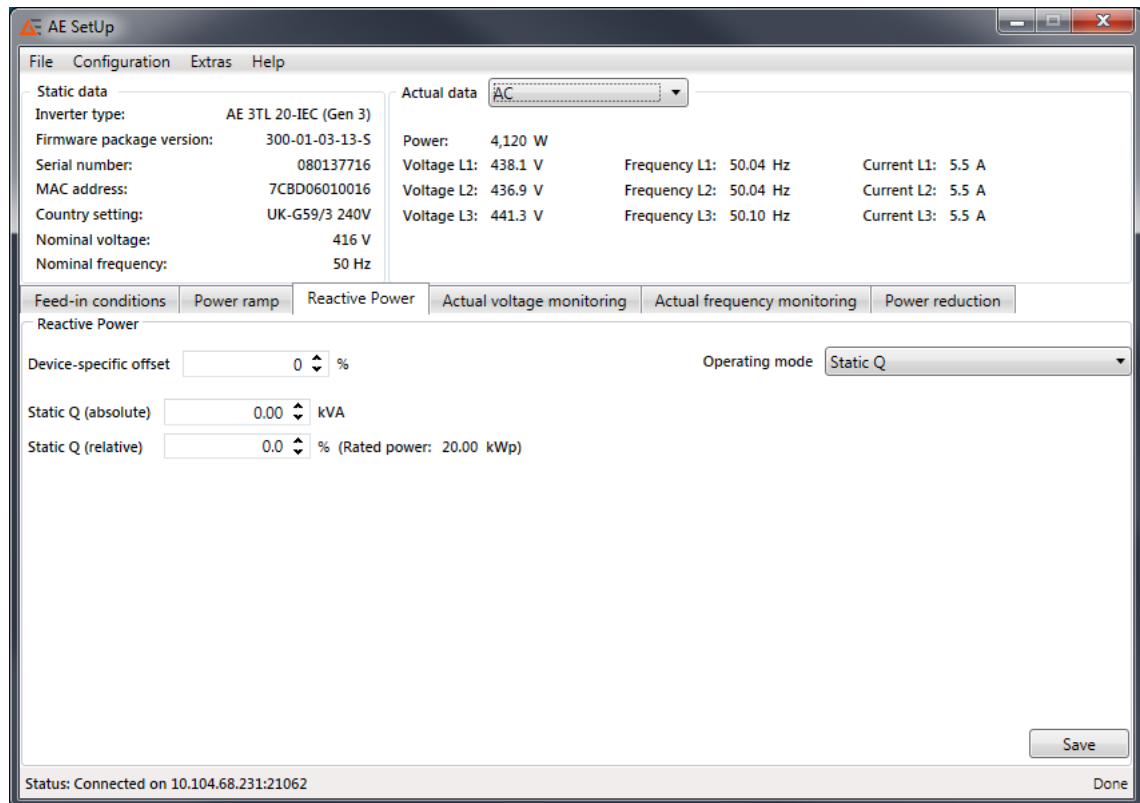


Fig. 21: Static value for reactive power

4.11.8. Q(U) characteristic

Depending on the voltage, a characteristic curve for the reactive power can be defined. In addition, you can enter an activation, deactivation threshold and the system-specific offset for power supply, for the hysteresis you can also define characteristic points. The resulting curve can be displayed as a graph. As a characteristic type are "linear interpolation" and "Hysteresis" available.

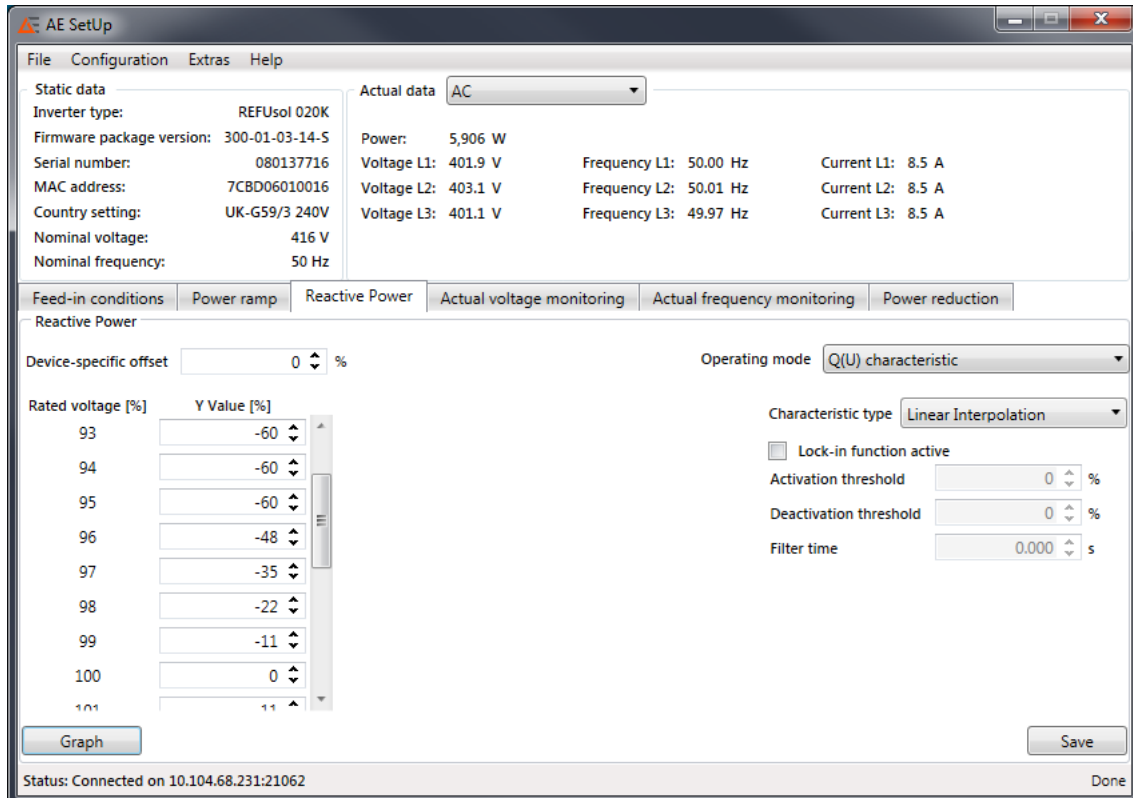


Fig. 22: Determining Reactive Power characteristic depending on voltage

You can display the characteristic entered in a chart by pressing the "Graph" button.

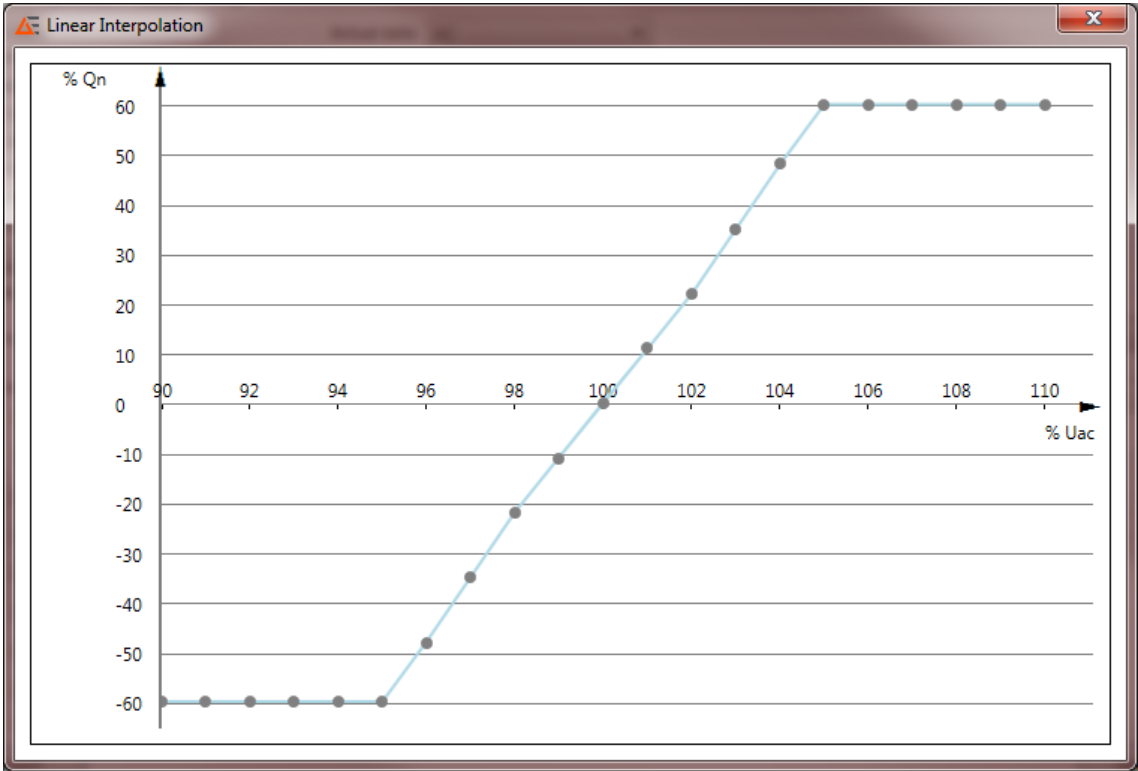


Fig. 23: Chart Characteristic type "linear interpolation"

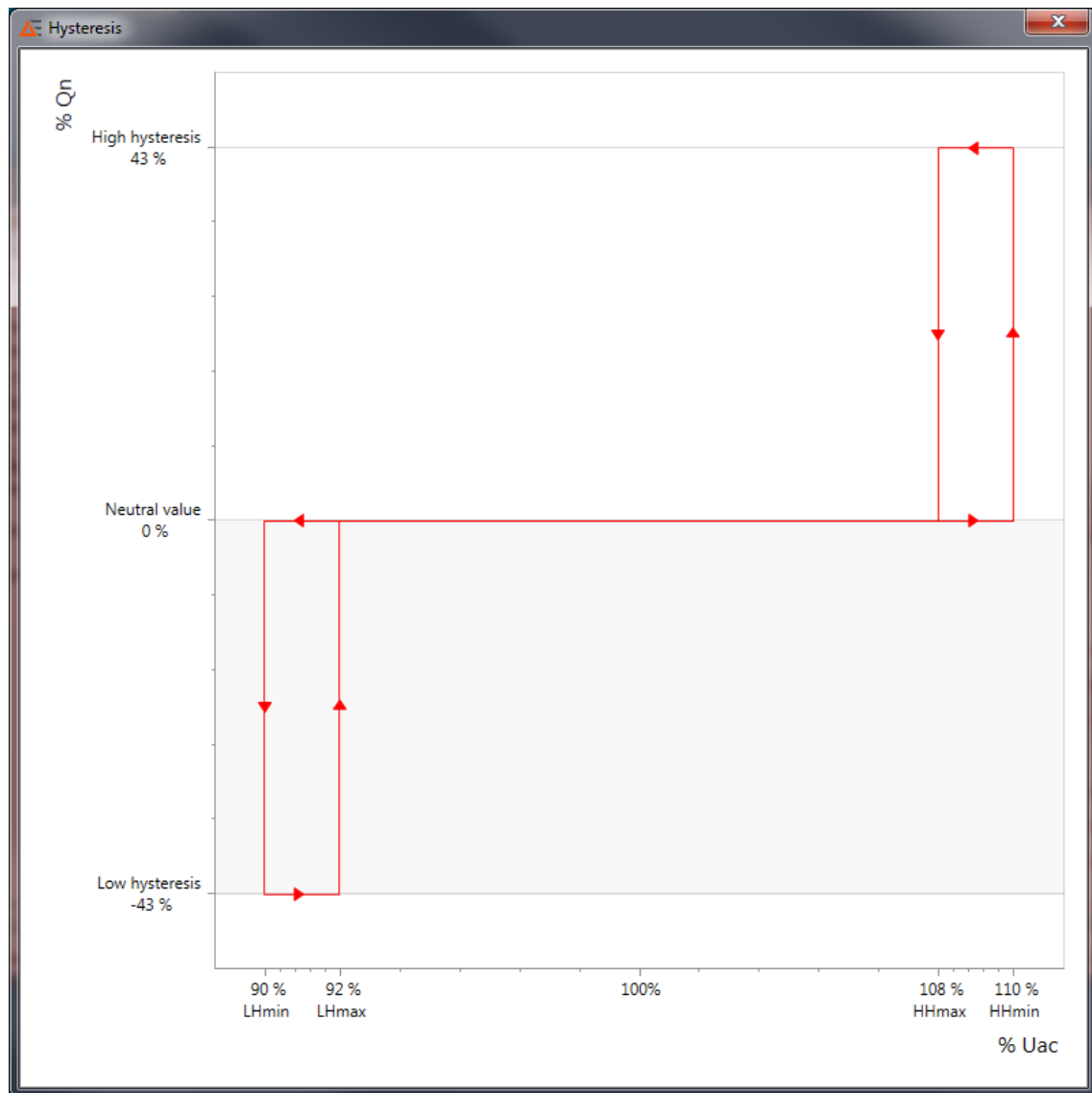


Fig. 24: Chart Characteristic type "Hysteresis"

4.11.9. Variable Q with reduced active power

The Q value is determined by other external, compatible regulation and control units. In this screen only the device-specific offset can be adapted.

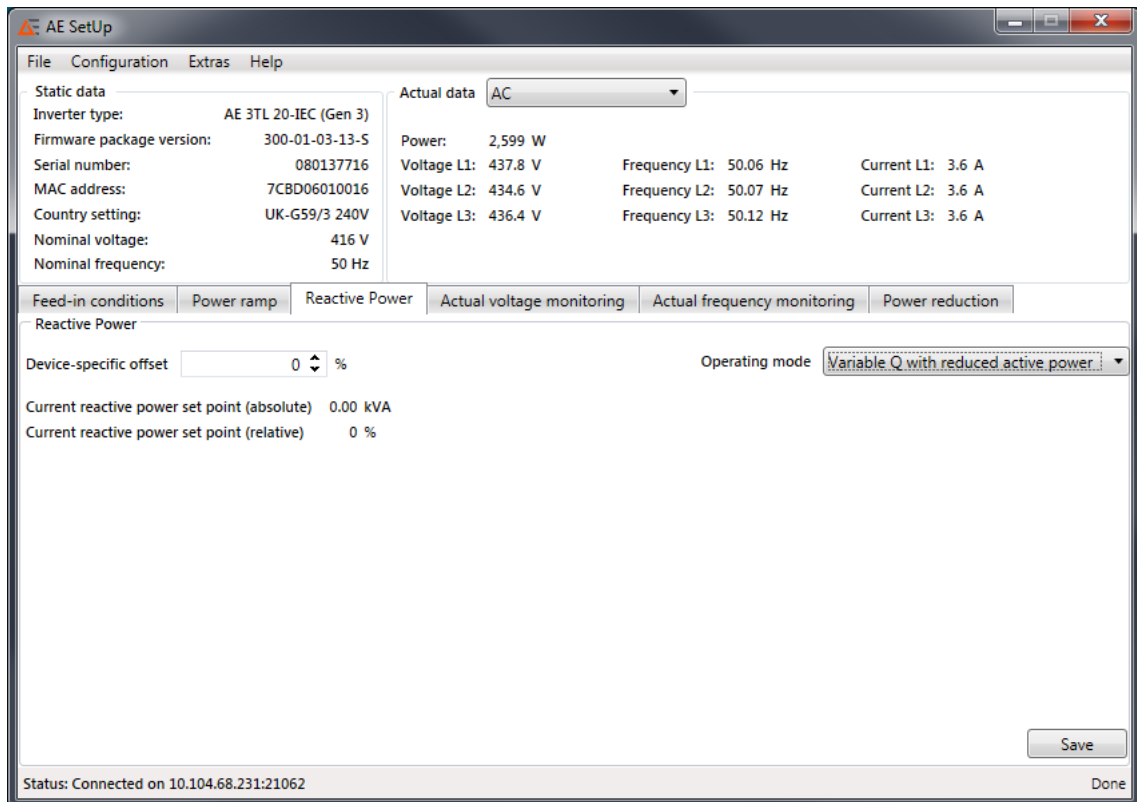


Fig 25: Variable Q with reduced power

4.11.10. Static Q with reduced active power

It is possible to enter a Q-fixed value. This can be expressed in absolute terms as Q-fixed value in kVA or relative in %. The conversion from absolute to relative or vice versa is done automatically. In addition, you can change the device-specific offset.

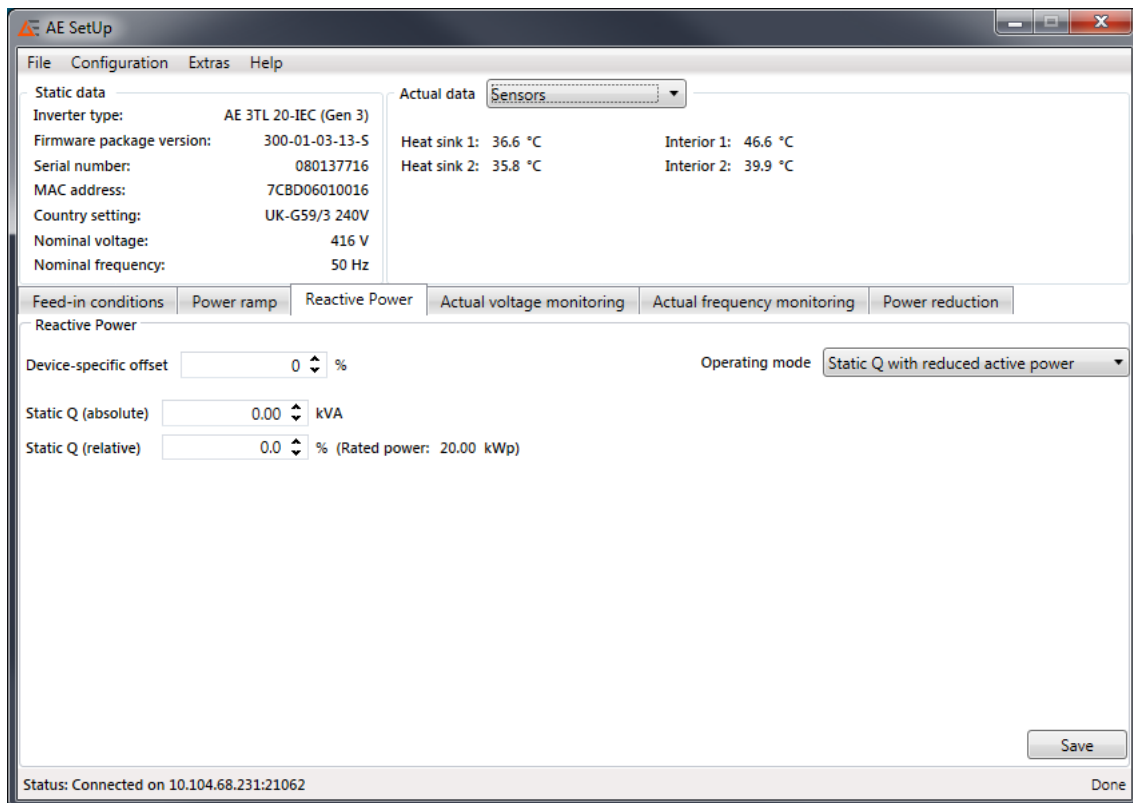


Fig. 26: Q fixed value with reduced power

4.12. FILE

In the **File** dropdown menu, you can establish a connection without closing the program, create a new connection or terminate the program.

As described above, a password for changing the country-specific settings can be set. In addition, a language can be set and the Anglo-American or Metric unit systems can be selected.

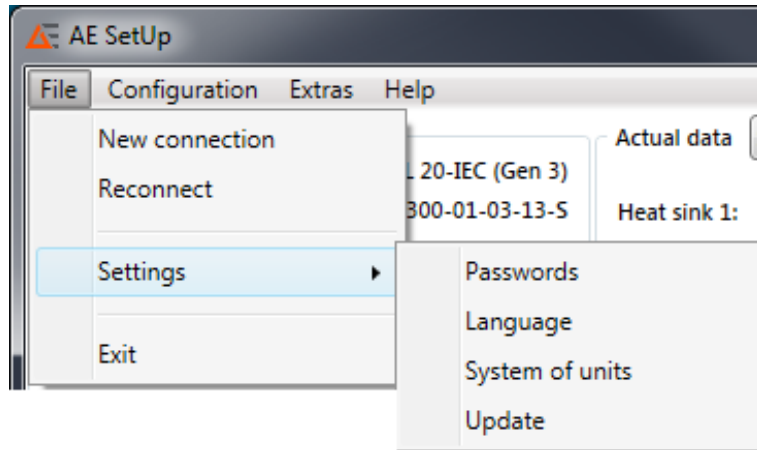


Fig. 27: File Dropdown Menu

4.13. CONFIGURATION

You can access the following subitems via the Inverter configuration menu item: **Communication**, **Data logger**, **Analog input configuration**, **AE SiteLink** and **Set time to computer time**.

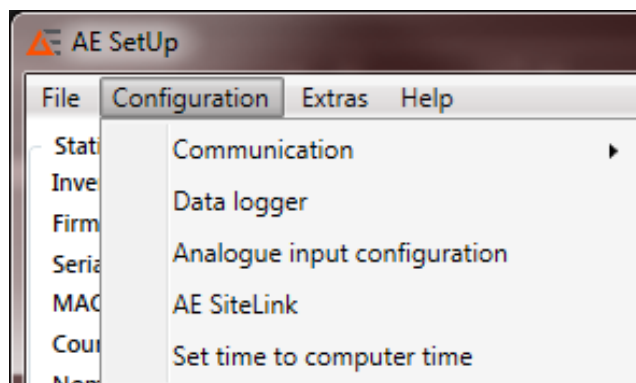


Fig. 28: Inverter Configuration Dropdown Menu

4.13.1. Ethernet-Configuration

In the menu item **Communication** → **Ethernet-Configuration** you can see the IP-Configuration of the inverter and change it.



Hinweis

IP configuration and RS485 settings will not be effective until after re-starting. You can only restart directly via the inverter thus far.

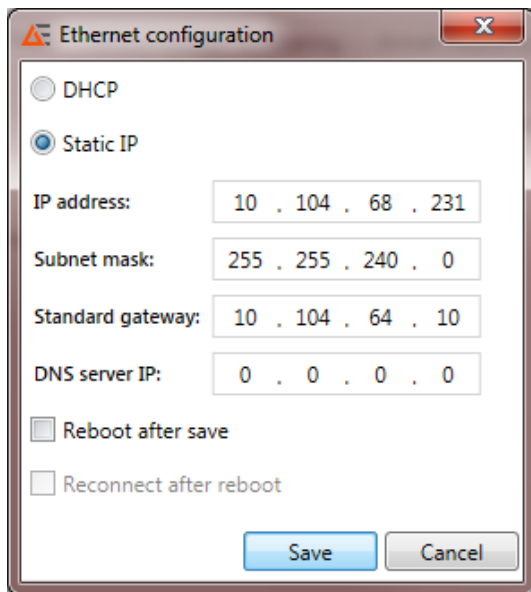


Fig. 29: Connection Settings

4.13.2. RS485

You can set the baud rate and the USS address in the **Communication** → **RS485** menu item.

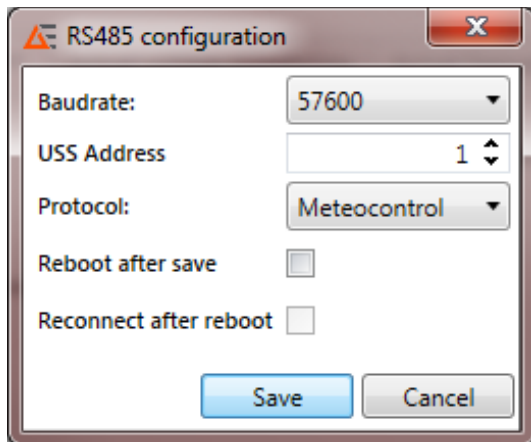


Fig. 30: RS485 Settings

4.13.3. Data logger

You can set the data logger's time interval in the **Configuration** → **Data logger** menu item.

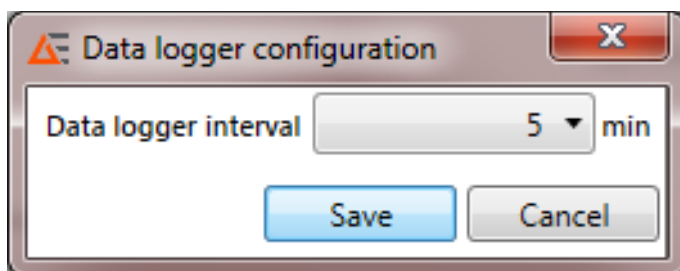


Fig. 31: Data Logger Configuration

4.13.4. Analog Inputs

You can configure analog sensor inputs in the **Configuration → Analog Inputs** menu item. This is the insolation sensor and the temperature sensor.

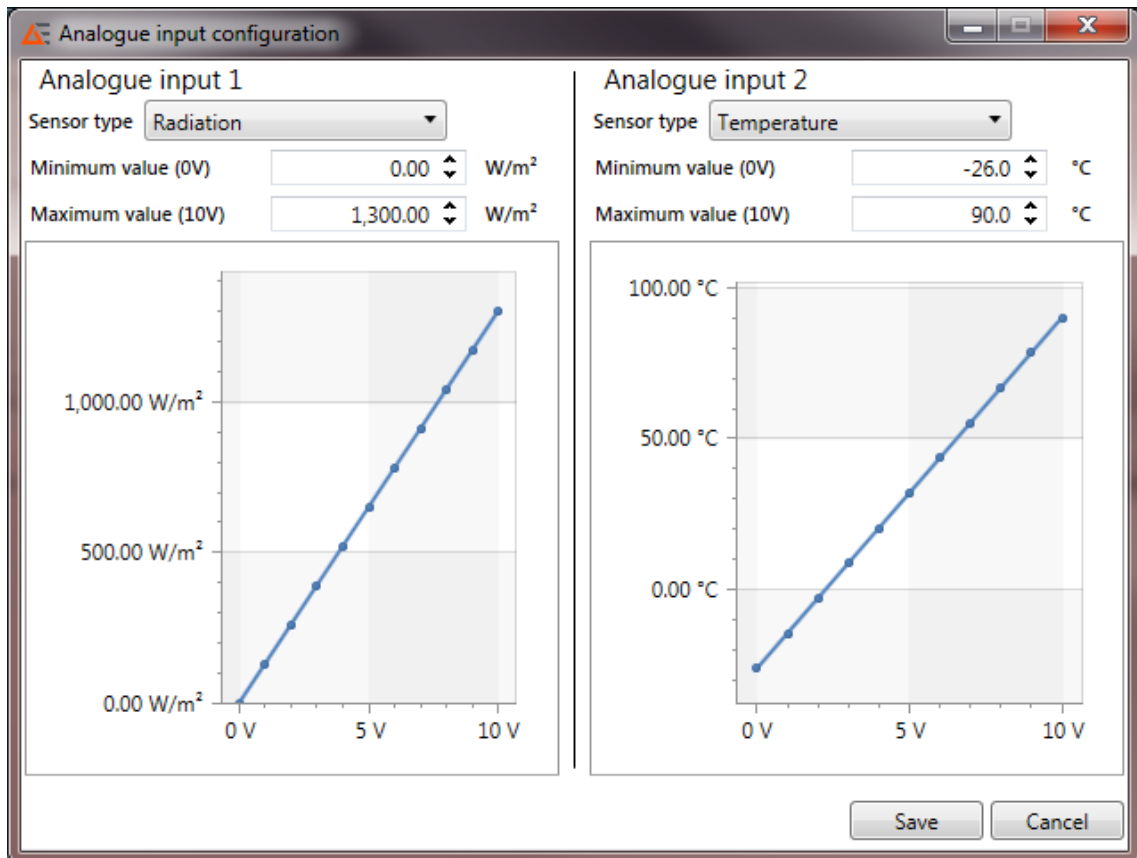


Fig. 32: Radiation and Temperature

If you have connected an external monitor to one of the analog inputs, please select the sensor type "External monitoring". It can be configured in the mask displayed below.

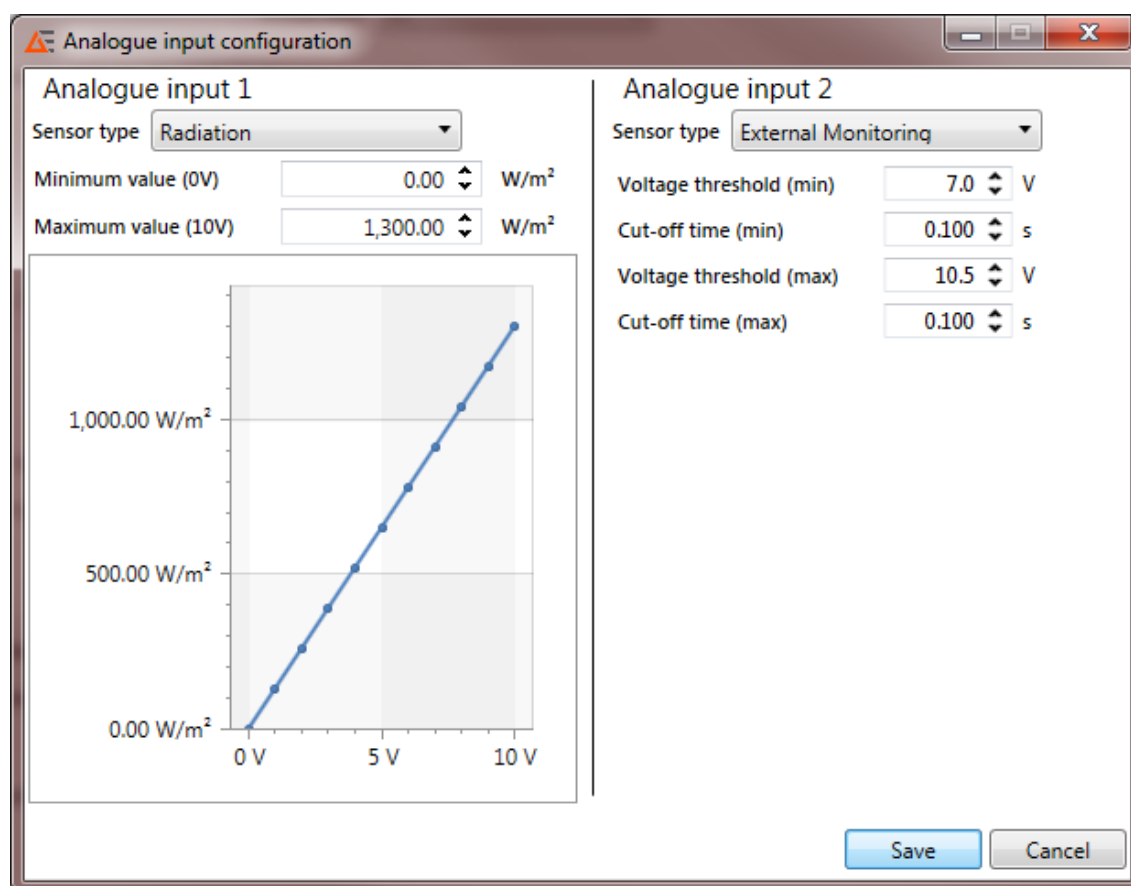


Fig. 33: Radiation and Temperature

4.13.5. AE SiteLink

You can activate the portal data forwarding function in the **Configuration → AE SiteLink** menu item. In addition, a test function and the configuration of the inverter can be sent from here.

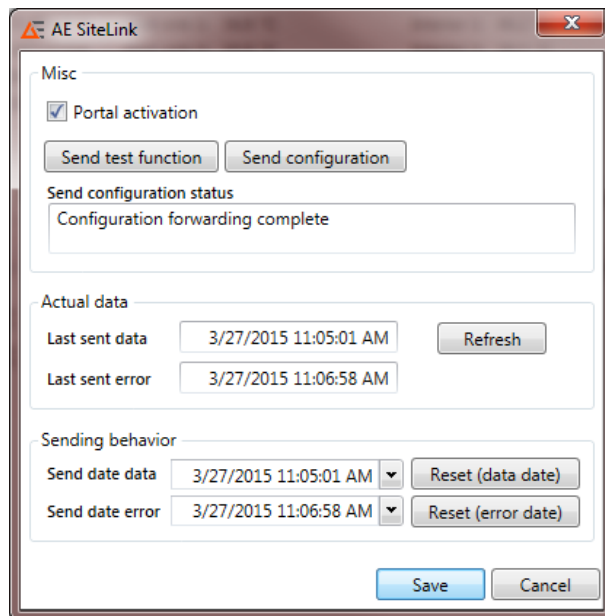


Fig. 34: AE SiteLink configuration

The date of the last data shipment and delivery of the last error is in the "Current data" visible.

In the area "sending behavior" you can enter a past date from that the data will be transmitted again. By pressing the button "Reset" all data is sent from the beginning of the recording.

4.13.6. Set time to computer time

By pressing configuration **Configuration → Set time to computer time**, the inverter time will be synchronized to the computer time. The following acknowledgment message appears.

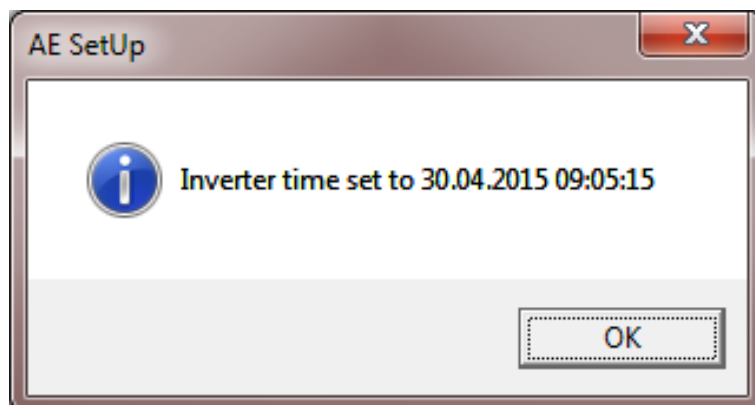


Fig. 35: Message after setting the system time

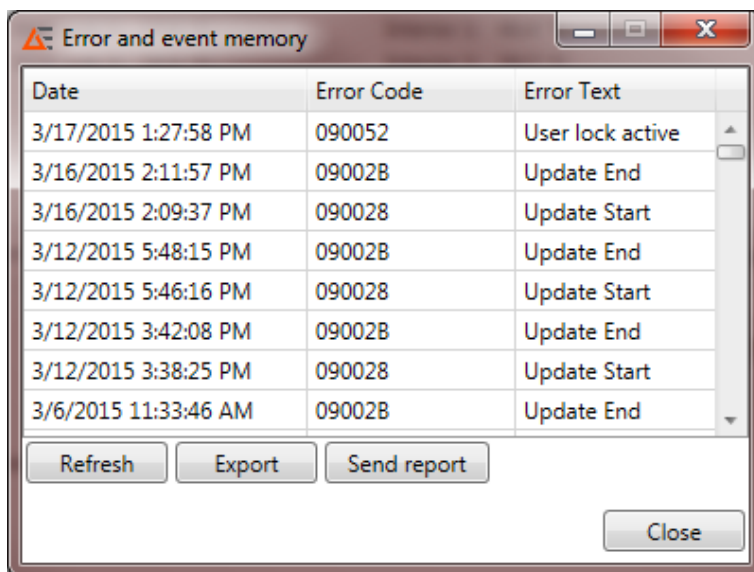
4.14. EXTRAS

4.14.1. Reboot

You can do a remote controlled restart with Extras → Reboot.

4.14.2. Error and event memory

Clicking the button "Extras Error/event memory" produces on the screen the following list:



Date	Error Code	Error Text
3/17/2015 1:27:58 PM	090052	User lock active
3/16/2015 2:11:57 PM	09002B	Update End
3/16/2015 2:09:37 PM	090028	Update Start
3/12/2015 5:48:15 PM	09002B	Update End
3/12/2015 5:46:16 PM	090028	Update Start
3/12/2015 3:42:08 PM	09002B	Update End
3/12/2015 3:38:25 PM	090028	Update Start
3/6/2015 11:33:46 AM	09002B	Update End

Buttons: Refresh, Export, Send report, Close

Fig. 36: Error and event memory

By pressing the appropriate buttons, it is possible to update this list, to generate a XML-file or to send the report by E-mail to AE.

After pressing "Export Parameters", the following screen will appear:

4.14.3. Firmware-Update

 CAUTION	Material damage during interrupt the update process possible. ⇒ Only perform update when sufficient sunlight is available until the end of the process.
---	--



Note

It is not possible to install an older firmware version. If this is necessary, contact the service.

For a firmware update, proceed as follows:

1. Download current firmware package on AE download and save it local on your computer.

2. About button "Select file" find corresponding .rfwps file.
3. Click Button „Update“.
 - ⇒ Update process is displayed on the progress bar.
 - ⇒ After successful upgrade, a status message appears.

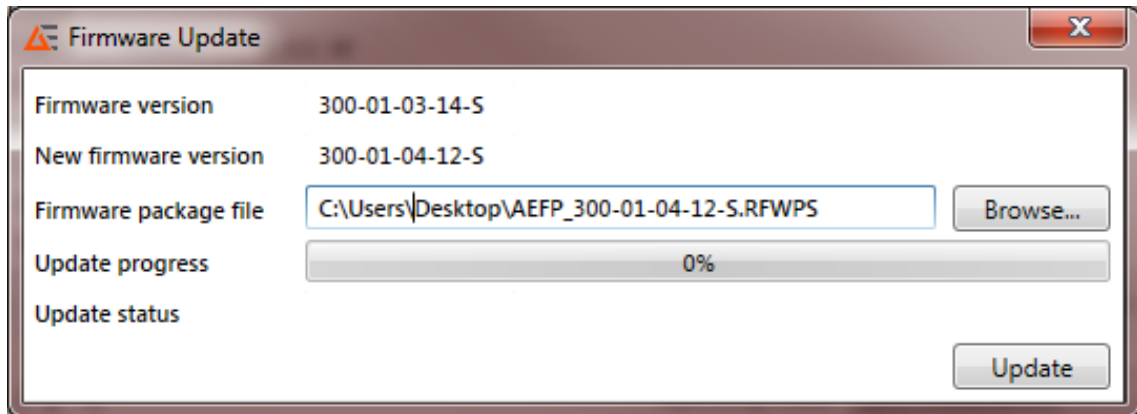


Fig. 37: Firmware-Update

4.15. HELP

You will see information about the tool's product version by clicking on the Help → About button.

Product version

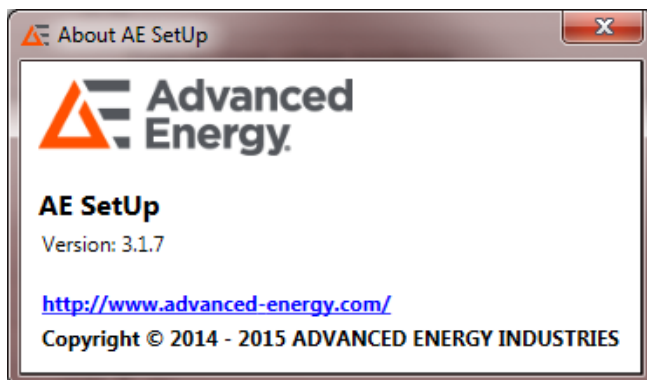


Fig. 38: Version status

Check for updates

With Help → Check for updates you can find out, whether the newest version of AE SetUp is installed.